

Hence work done is the area under E-field distance graph.

20 A

$$\text{Power in } R \text{ in circuit (i)} = I^2 R = \left(\frac{3E}{3r+R} \right)^2 R = \left(\frac{3(1.5)}{3 \times 2 + 4} \right)^2 \times 4 = 0.81 \text{ W}$$

$$\text{Power in } R \text{ in circuit (ii)} = I^2 R = \left(\frac{E}{3r+R} \right)^2 R = \left(\frac{1.5}{3 \times 2 + 4} \right)^2 \times 4 = 0.09 \text{ W}$$

$$\frac{\text{power in } R \text{ in circuit (i)}}{\text{power in } R \text{ in circuit (ii)}} = \frac{0.81}{0.09} = 9$$

21 C